

HW4 Prob. 3

3. a) Find max height, $x = \mu$, probability

$$h = \frac{1}{\sigma\sqrt{2\pi}} e^{\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]} = \frac{1}{\sigma\sqrt{2\pi}} e^0 = \frac{1}{\sigma\sqrt{2\pi}}$$

half-max : $\frac{h}{2} = \frac{1}{2\sigma\sqrt{2\pi}} \rightarrow$ find x pos. where this is true:

$$\frac{1}{\cancel{\sigma\sqrt{2\pi}}} e^{\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]} = \frac{1}{2\cancel{\sigma\sqrt{2\pi}}}$$

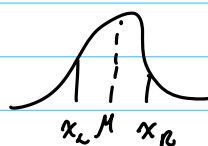
$$\ln\left(e^{\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]}\right) = \ln\left(\frac{1}{2}\right)$$

$$-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2 = \ln\left(\frac{1}{2}\right)$$

$$\frac{x-\mu}{\sigma} = \pm\sqrt{2\ln(2)}$$

$x = \mu \pm \sigma\sqrt{2\ln(2)} \rightarrow$ this will be to the right & left of μ

Full width: distance between:



$$d = x_R - x_L$$

$$= \cancel{\mu} + \sigma\sqrt{2\ln(2)} + (\cancel{-\mu} + \sigma\sqrt{2\ln(2)})$$

$$= 2\sqrt{2\ln(2)}\sigma \approx 2.354\sigma \text{ as expected!}$$

b. log-logs have $\log(y)$ on y-axis & $\log(x)$ on the x-axis.

$\rightarrow$ a line on log-log : $\log(y) = m\log(x) + b$ let's say $b = \log(a)$
both a & b are constants

$$\log(y) = m\log(x) + \log(a)$$

$$\log(y) = \log(x^m(a))$$

b) cont.

$$\log_{10}(y) = \log_{10}(x^m (a))$$

$$y = a x^m$$

thus b , the intercept of the log-log graph represents $\log$ of the coeff. of x and the slope, m , of the log-log graph represents the power of x